Euclid, Book I, Proposition I.9

To Bisect a Given Rectilinear Angle

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1 Introduction

Proposition I.9 presents Euclid's construction of the bisector of a given rectilinear angle.

The classical proof combines several earlier propositions. A segment of one side of the angle is first copied onto the other side, after which an auxiliary equilateral triangle is constructed. The Side–Side–Side criterion established in Proposition I.8 is then used to prove that the joining segment bisects the angle.

The Hilbert reconstruction follows a different route. Rather than constructing an auxiliary equilateral triangle, it first lays off a segment congruent to one side of the angle, constructs the midpoint of the resulting segment, and finally applies the general Side–Side–Side theorem.

Although the geometric constructions differ substantially, both approaches establish the existence of an angle bisector within neutral geometry.

2 The Classical Configuration

Let $\angle BAC$ be the given rectilinear angle.

The objective is to construct a ray issuing from the vertex A that divides the angle into two congruent angles.

No assumption is made on the lengths of the sides forming the angle; only the angle itself is given.

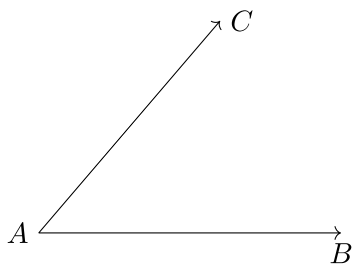

Figure 1: The given angle $\angle BAC$.

2.1 Construction

Choose an arbitrary point D on the side AB.

On the side AC, construct a point E such that

$$AE \cong AD.$$

Join the points D and E, and construct the equilateral triangle DEF on the segment DE.

Finally, join the vertex A to the vertex F.

The claim is that the segment AF bisects the given angle.

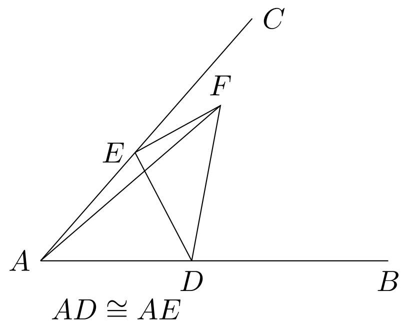

Figure 2: Euclid's auxiliary construction: $AD \cong AE$ and DEF is equilateral.

2.2 Application of Proposition I.8

Since the triangle DEF is equilateral,

$$DF \cong EF.$$

Moreover,

$$AD \cong AE$$

by construction, while

$$AF \cong AF$$

is common to the two triangles.

Therefore the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Since D lies on the ray AB and E lies on the ray AC, these are precisely the two parts of the original angle $\angle BAC$.

Hence AF bisects the given angle.

3 Statement

Theorem 1 (Euclid I.9)**.** *Let $\angle BAC$ be a nondegenerate angle.*

Then there exists a point M such that

$$\angle BAM \cong \angle MAC.$$

Equivalently, the ray AM bisects the angle $\angle BAC$.

4 Classical Proof

Choose a point D on the side AB and construct a point E on the side AC such that

$$AD \cong AE.$$

Join the points D and E, and construct the equilateral triangle DEF.

Finally, join the vertex A to the vertex F.

Since

$$AD \cong AE,$$

$$DF \cong EF,$$

and

$$AF \cong AF,$$

the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Because D lies on the ray AB and E lies on the ray AC, these angles are exactly

$$\angle BAF \quad \text{and} \quad \angle FAC.$$

Hence the ray AF bisects the given angle.

This completes the construction.

5 Proof Architecture of the Reconstruction

The Hilbert reconstruction replaces Euclid's auxiliary equilateral triangle by two fundamental constructions already available in the Hilbert interface.

First, a segment congruent to one side of the angle is laid off on the other side of the angle. The midpoint of the resulting segment is then constructed, and the Side–Side–Side theorem is applied to two auxiliary triangles.

Thus the existence of the angle bisector follows entirely from previously established neutral-geometric results.

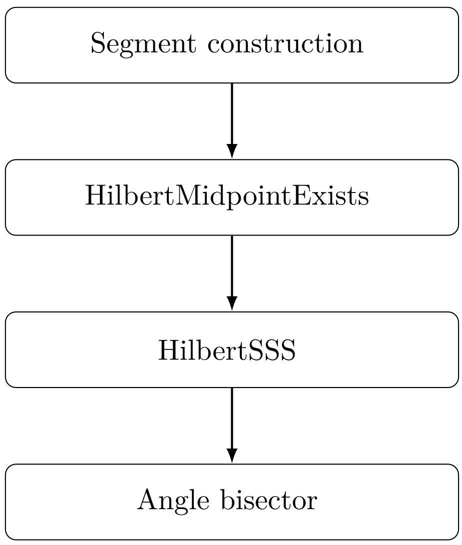

Figure 3: Architecture of the Hilbert reconstruction of Proposition I.9.

6 Hilbert Reconstruction

Starting from the given angle, a point is first constructed on one ray so that the new segment is congruent to the side on the opposite ray.

The midpoint of the segment joining these two constructed endpoints is then obtained using the neutral midpoint theorem.

Finally, the triangles determined by the midpoint satisfy the hypotheses of the general Side–Side–Side theorem. The resulting angle congruence shows that the segment joining the vertex of the angle to the midpoint is an angle bisector.

The proof therefore avoids both the construction of an equilateral triangle and the explicit application of Euclid's Proposition I.8, replacing them by previously established interface theorems.

7 Lean Formalization

The Lean implementation follows the Hilbert reconstruction directly.

Starting from a nondegenerate angle, the proof first constructs a point on one ray whose distance from the vertex equals the length of the opposite side of the angle.

The midpoint of the segment joining the two constructed endpoints is then obtained using the neutral midpoint existence theorem.

Finally, a single application of the general Side–Side–Side theorem establishes the congruence of the two auxiliary triangles. The desired angle congruence follows immediately after identifying the constructed segment with the original ray.

Unlike Euclid's original argument, the formal proof contains no equilateral triangle construction and makes no direct use of Proposition I.8. Instead, it relies entirely on previously established results of the Hilbert interface.

8 Relationship with Earlier Propositions

8.1 Proposition I.1

Euclid begins the construction by choosing points on the two arms of the given angle and constructing an equilateral triangle on the segment joining them. Proposition I.1 supplies the symmetric auxiliary configuration from which the angle bisector will be obtained.

8.2 Proposition I.3

The classical proof uses segment cutting to arrange equal auxiliary lengths on the two sides of the given angle. Proposition I.3 supplies this segment-layoff step.

8.3 Proposition I.8

Proposition I.8 is the decisive congruence theorem in Euclid's proof. The two triangles adjacent to the candidate bisector have three corresponding sides congruent, so SSS yields congruence of the two angles at the original vertex.

Thus the classical dependency pattern is

$$\mathrm{I.3} \longrightarrow \mathrm{I.1} \longrightarrow \mathrm{I.8} \longrightarrow \mathrm{I.9}.$$

8.4 Transition to Proposition I.10

Proposition I.10 uses the angle-bisection construction immediately: an equilateral triangle is erected on the given segment and its apex angle is bisected. SAS then shows that the point where the bisector meets the base is the midpoint.

Thus I.9 becomes constructional infrastructure as soon as it is proved.

9 Hilbert and Book Zero Components Used

9.1 Angle-Bisector Representation

The conclusion is expressed synthetically by a ray from the vertex whose two adjacent angles are congruent. No numerical angle measure is used.

9.2 Segment Layoff

The Hilbert segment-construction machinery supplies the equal auxiliary segments needed to create the symmetric configuration around the given angle.

9.3 SSS

The congruence of the two auxiliary triangles is obtained from the established Hilbert SSS theorem. The desired angle congruence is then a corresponding-angle consequence.

9.4 Interior and Ray Information

The constructed bisector must lie in the intended interior region of the given angle. The formal proof therefore carries ray and side-of-line information that is implicit in Euclid's diagram.

10 Formalization Notes

Proposition I.9 is the first construction in Book I whose Hilbert reconstruction differs substantially from Euclid's original proof.

Euclid constructs an auxiliary equilateral triangle and applies the Side–Side–Side criterion established in Proposition I.8. The proof is therefore closely tied to the historical sequence of propositions in the *Elements*.

The Hilbert reconstruction adopts a different strategy. A congruent segment is first laid off on one side of the angle, the midpoint of the resulting segment is constructed, and the general Side–Side–Side theorem is applied directly to two auxiliary triangles.

This approach illustrates a general principle of the CGJteamLab library: classical geometric constructions are not reproduced verbatim when a simpler synthetic argument is available within the existing Hilbert theory.

Proposition I.9 therefore demonstrates how a modern axiomatic development can replace a classical construction by a shorter proof while preserving the same mathematical content.

11 Dependency Summary

The formal reconstruction follows the same basic synthetic geometry as the classical proof:

$$\begin{array}{c}\text{nondegenerate angle}\\ \downarrow\\ \text{lay off equal auxiliary segments}\\ \downarrow\\ \text{construct symmetric auxiliary triangle}\\ \downarrow\\ \text{SSS}\\ \downarrow\\ \text{congruent adjacent angles}\\ \downarrow\\ \text{angle bisector}\\ \downarrow\\ \texttt{euclid\_proposition\_9}.\end{array}$$

The reconstruction uses only the established Hilbert and Book Zero infrastructure. It introduces neither numerical angle measure nor a new geometric axiom.

12 Conclusion

Proposition I.9 constructs the bisector of a given rectilinear angle.

The classical proof is an elegant combination of the early construction theorems and SSS: equal auxiliary segments create a symmetric configuration, and triangle congruence converts that symmetry into equality of the two angles at the vertex.

The formal reconstruction preserves this synthetic character while making the positional content explicit. A bisector is not merely an abstract equality of two angle magnitudes; it is represented by a specific ray lying in the appropriate geometric configuration.

I.9 is also an important infrastructural milestone. The construction is used immediately in Proposition I.10 and becomes part of the reusable angle-construction language needed throughout the later development.